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SUPPLEMENTARY INFORMATION

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The short-term impacts of COVID-19 lockdown on urban air pollution in China

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Supplementary Information for
"The Short-term Impacts of COVID-19 Lockdown on Urban Air Pollution in China"

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Supplementary Notes

Supplementary Note 1: The Trends in Air Quality

We present the patterns in the raw air quality data. In Panel A of Supplementary Figure 1, we plot the Air Quality Index (AQI) between the treatment (locked-down cities) and control cities (non-locked-down cities) over the study time in 2020. This figure shows to what extent city lockdown affected air pollution between the treated and control cities. The treatment group had worse air pollution levels (higher AQI) than the control group at the beginning of January. However, the difference significantly decreased after more cities were locked down, suggesting that the lockdown improved air quality.

In Panel B, we investigate to what extent the control cities were also affected by the counter-virus measures. We see the AQI levels were almost equivalent before the start of the Spring Festival holiday in 2019 and 2020. In 2020, shortly after the start of the festival, we observe that the air pollution levels became slightly lower, relative to 2019. This result suggests that air quality in the control cities marginally improved, although they were not formally locked down.

Supplementary Note 2: The Effect of City Lockdown in Pre-Spring Festival

We estimate equation (1) using data before the Spring Festival and summarize the results in Supplementary Table 9. Before the Spring Festival, only Wuhan and a few neighboring cities enforced lockdown policies, and most other cities in China had not yet adopted any counter-virus measures. As a result, using this restricted sample gives us a relatively "clean" control group, which may help us capture the overall impact of city lockdowns.

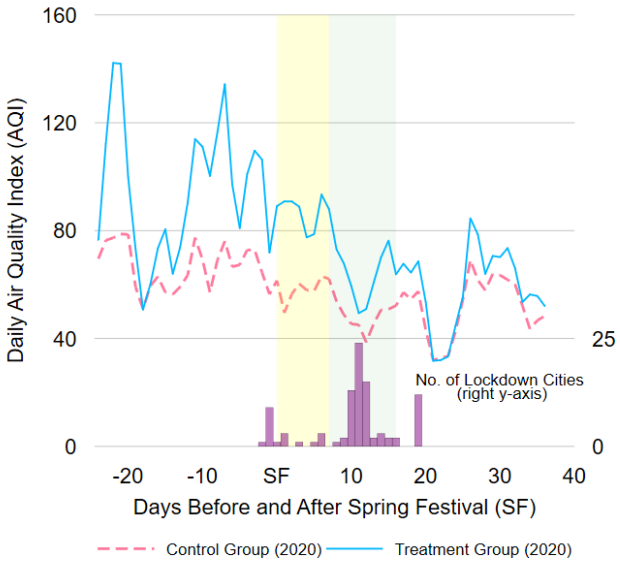
One may expect the estimated impact of lockdowns on air quality would be larger using this restricted sample because cities in the control group did not have any counter-virus measure. However, as shown in Supplementary Table 9, the results are almost identical to our baseline estimates.

There are two reasons why we do not observe a larger impact using the restricted sample. First, it takes time for air pollutants to be dispersed or settle down. In fact, our event study shows that the lockdown effect becomes larger as more lags are included (Figure 3). When we use the restricted sample, we only include data several days after the lockdowns into the regression. As we are unable to capture the accumulated effect using the restricted sample, it is unsurprising to see a lower-than-expected effect. Second, in practice, as more COVID-19 cases were identified, local governments could also gradually tighten the counter-virus measures. This would also lead to a smaller lockdown effect at the beginning.

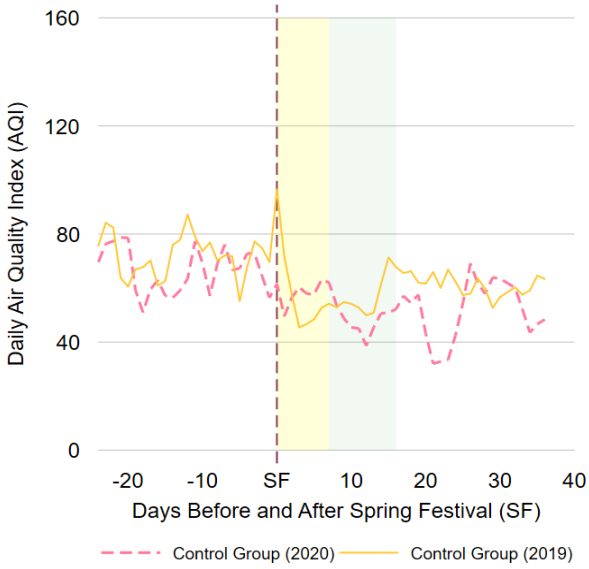
Supplementary Figures

Supplementary Figure 1: Trends in Air Quality in the Treatment and Control Groups

Panel A: Trend in AQI in the Treatment and Control Group in 2020



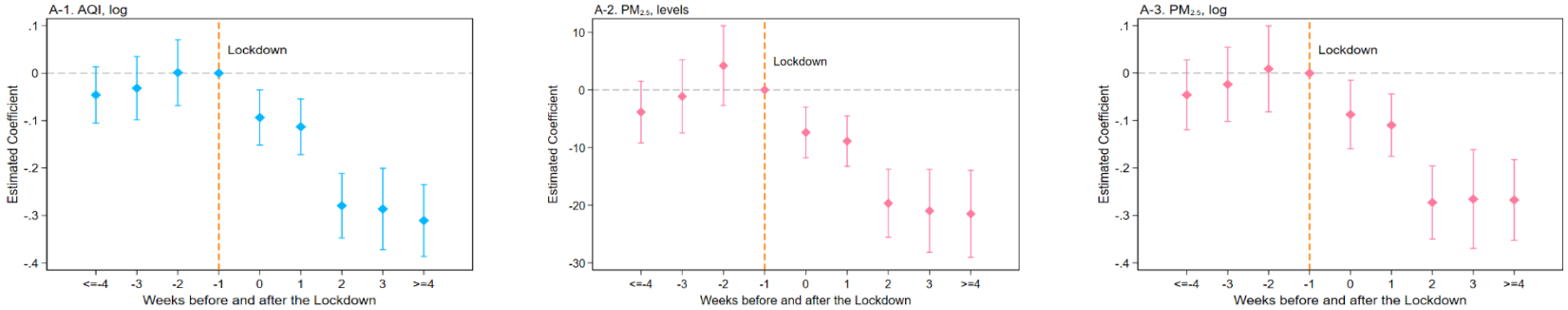
Panel B: Trend in AQI in the Control Group in 2019 and 2020



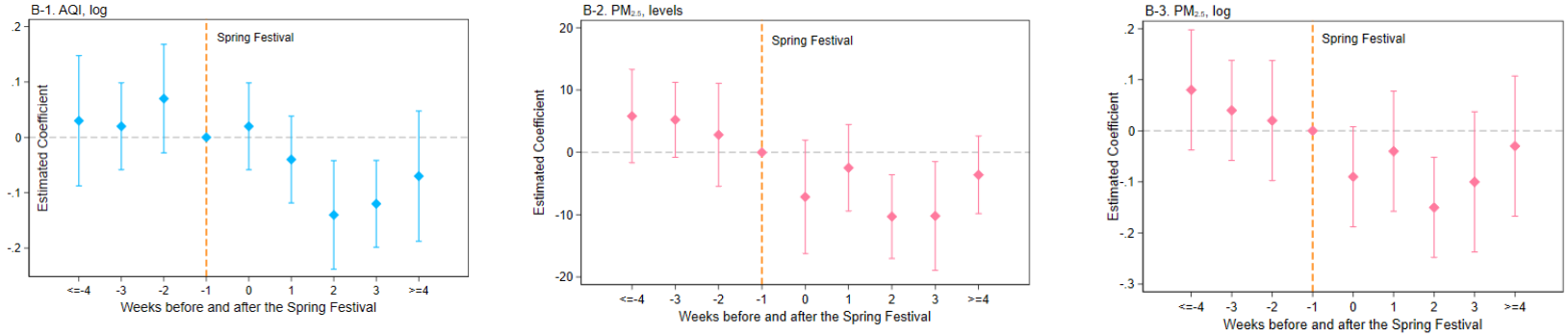
Notes: These figures show the trends in the Air Quality Index (AQI) in different groups of cities. In Panel A, we plot the AQI in the treatment and the control group in 2020. In Panel B, we plot the AQI in the control group in 2019 and 2020. The vertical dashed purple line represents the timing of the Chinese Spring Festival. The light-yellow shadow indicates the Spring Festival Holiday period, and the light-green shadow shows the expanded national holiday in 2020 (January 31st ~ February 10th).

Supplementary Figure 2. Event-Study Results on AQI and $PM_{2.5}$

Panel A. The Effects of Lockdown on Air Quality

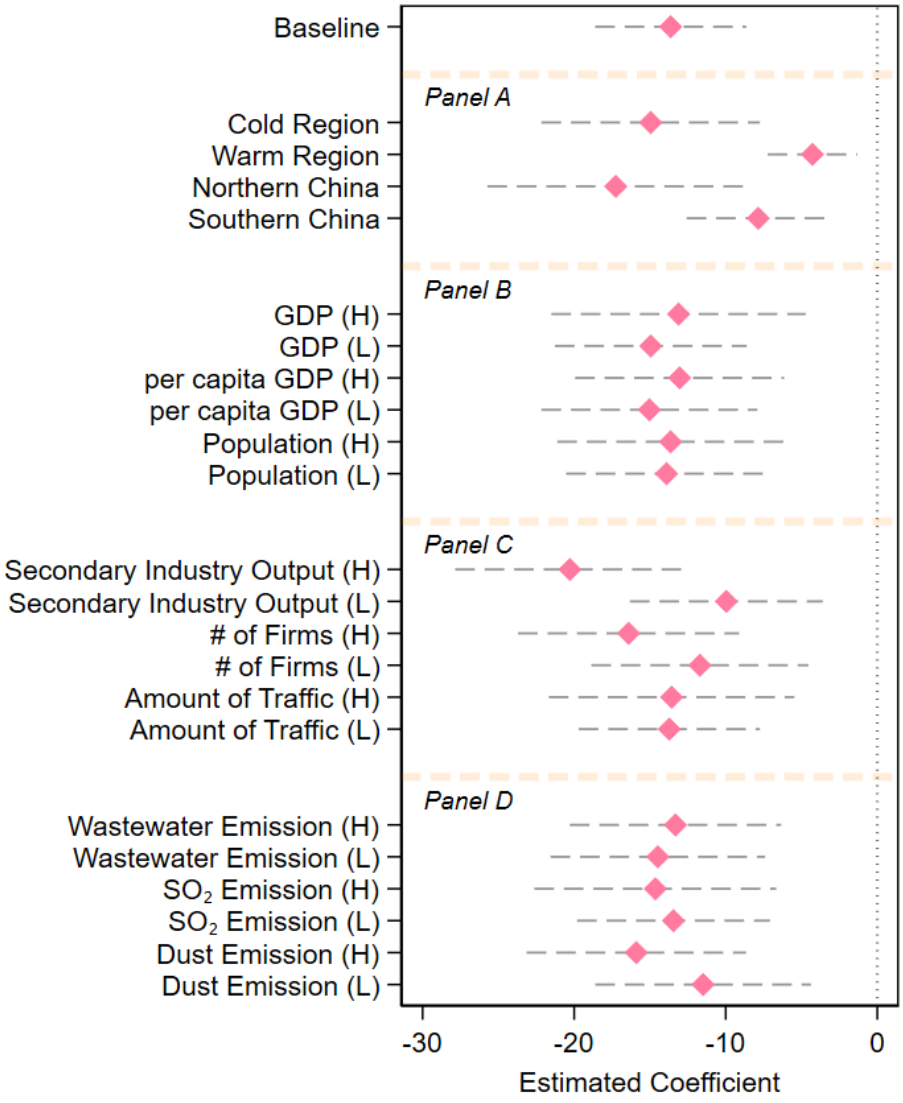


Panel B. The Effects of 2020 Spring Festival on Air Quality in the Control Group



Notes: These figures summarize the results of the event study. We include leads and lags of the start of the city lockdown dummy in the regressions. The dummy variable indicating one week before the city lockdown is omitted from the regressions. The estimated coefficients and their 95% confidence intervals are plotted. In Panel A, we compare the air pollution levels between the treated cities with the control cities, and the vertical line indicates the timing of lockdowns. In Panel B, we compare air pollution levels in the control cities between 2019 and 2020. The vertical line indicates the start of the Spring Festival holiday.

Supplementary Figure 3. Heterogeneous Impact Using $PM_{2.5}$



Notes: The X-axis shows the estimated coefficients and their 95% confidence intervals. Each row corresponds to a separate regression using a corresponding subsample. We use the mean values to separate the "high" group from the "low" group for each pair of heterogeneity analyses. For example, if a city's GDP is higher than the mean GDP, it falls into a "high" GDP group. For temperature (colder or warmer group), we use data measured in the first week of our study period. North and South are divided by the Huai River. Other socio-economic data for the classification are measured in 2017. Each regression implements the model (1) and controls for the weather, city fixed effects, and date fixed effects. Standard errors are clustered at the city level.

Supplementary Tables

Supplementary Table 1. The Relationship between the AQI and Air Pollutant Concentrations

AQI	Each Air Pollutants						Air Quality Levels
	PM_{10} (24hr)	$PM_{2.5}$ (24hr)	NO_2 (24hr)	O_3 (8hr)	CO (24hr)	SO_2 (24hr)	
0-50	0-50	0-35	0-40	0-100	0-2	0-50	Excellent
50-100	50-150	35-75	40-80	100-160	2-4	50-150	Good
100-200	150-350	75-150	80-280	160-265	4-24	150-800	Slightly Polluted
200-300	350-420	150-250	280-565	265-800	24-36	800-1600	Moderately Polluted
300-400	420-500	250-350	565-750	/	36-48	1600-2100	Severely Polluted
400-500	500-600	350-500	750-940	/	48-60	2100-2620	Severely Polluted

Notes: This table reports the AQI sub-index levels for each air pollutant. The sub-index with the highest value will then be used as the AQI. For CO, the unit is mg/m^3, and for other pollutants, the units are $\mu g/m^3$.

Supplementary Table 2. List of Locked-down Cities and The Major Events

Starting Date	Cities, and the Major Events *(*)*
20-Jan	*(*) The national government disclaimed "the virus can transmit from people to people."*
23-Jan	Wuhan
24-Jan	Huangshi, Shiyan, Yichang, Ezhou, Jingmen, Xiaogan, Huanggang, Xianning, Enshi
25-Jan	*(*) The start of the Chinese Spring Festival*
	Qinhuangdao
26-Jan	*(*) The extension of the Chinese Spring Festival was announced.*
	Xiangyang, Jingzhou, Xiantao
28-Jan	Tangshan
30-Jan	*(*) The last day of the original Chinese Spring Festival*
	Dongying
31-Jan	Chongqing, Yinchuan, Wuzhong
2-Feb	Wenzhou
3-Feb	Wuxi, Jining
4-Feb	Harbin, Nanjing, Xuzhou, Changzhou, Nantong, Hangzhou, Ningbo, Fuzhou, Jingdezhen, Zaozhuang, Linyi, Zhengzhou, Zhumadian
5-Feb	Shenyang, Dalian, Anshun, Fushun, Benxi, Dandong, Jinzhou, Fuxin, Liaoyang, Panjin, Tieling, Chaoyang, Huludao, Yangzhou, Hefei, Quanzhou, Nanchang, Jinan, Qingdao, Taian, Rizhao, Laiwu, Nanning
6-Feb	Tianjin, Shijiazhuang, Suzhou, Pingxiang, Jiujiang, Xinyu, Yingtan, Ganzhou, Ji'an, Yichun, Fuzhou, Shangrao, Neijiang, Yibin, Xinyang
7-Feb	Suzhou, Guangzhou
8-Feb	Shenzhen, Foshan, Fangchenggang,
9-Feb	Cangzhou, Huaibei
10-Feb	*(*) The last day of the extended Chinese Spring Festival*
	Beijing, Shanghai
13-Feb	Hohhot, Baotou, Wuhai, Chifeng, Tongliao, Ordos, Hulun Buir, Bayan Nur, Ulanqab, Xing'an League, Xilingol League, Alxa League

Notes: The lockdown information is from local government and various media news in 2020.

Supplementary Table 3. Summary Statistics

	2020 (01. January ~ 01. March)				2019 (12. January ~ 13. March)	
		Treatment Group (Locked-down Cities)				
	All Cities	Before Treatment	After Treatment	Control Group	Treatment Group	Control Group
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Air Pollutant and Lockdown						
Air Quality Index (AQI)	75.01	102.54	69.60	70.37	100.49	76.67
	(52.13)	(63.92)	(37.86)	(49.97)	(57.53)	(48.54)
$PM_{2.5}$ ($\mu g/m^3$)	52.14	76.17	48.27	47.92	71.55	50.03
	(41.49)	(52.90)	(30.55)	(38.85)	(47.75)	(39.164)
Lockdown	0.14	0.00	1.00	0.00	/	/
	(0.35)	(0.00)	(0.00)	(0.00)	/	/
Panel B. Weather						
Temperature (℃)	3.60	2.47	6.14	3.34	3.26	2.86
	(8.44)	(6.45)	(5.97)	(9.10)	(5.65)	(7.85)
Precipitation (100mm)	24.22	20.18	17.84	26.30	40.11	47.40
	(65.08)	(46.38)	(44.38)	(71.40)	(44.48)	(74.74)
Snow Depth (100mm)	60.41	59.92	58.57	60.87	62.14	66.88
	(32.21)	(29.19)	(25.37)	(33.94)	(60.98)	(64.50)
Number of Cities	324	95	95	229	95	229

Notes: Each Column summarizes the mean values and standard deviations of different variables at the daily level.

Supplementary Table 4. Main Specification Using Other Air Pollutants

	Treatment and Control Cities in 2020				
	CO	NO_2	PM_{10}	SO_2	O_3
	(mg/m^3)	($\mu g/m^3$)	($\mu g/m^3$)	($\mu g/m^3$)	($\mu g/m^3$)
	(1)	(2)	(3)	(4)	(5)
Panel A. Levels					
Lockdown	-0.08***	-5.05***	-22.25***	-1.14**	4.53***
	(0.03)	(0.71)	(3.28)	(0.53)	(0.76)
R-Squared	0.607	0.745	0.496	0.707	0.546
Panel B. log					
Lockdown	-0.03***	-0.14***	-0.20***	-0.07***	0.13***
	(0.01)	(0.02)	(0.03)	(0.02)	(0.02)
R-Squared	0.638	0.798	0.645	0.803	0.542
Weather Control	Y	Y	Y	Y	Y
City FE	Y	Y	Y	Y	Y
Date FE	Y	Y	Y	Y	Y
Obs.	19,764	19,764	19,764	19,764	19,764
No. of Cities	324	324	324	324	324

Notes: Weather controls include daily temperature, its square, precipitation, and snow depth. Standard errors are clustered at the city level and reported below the coefficients. * significant at 10% ** significant at 5%. *** significant at 1%.

Supplementary Table 5. Event Study: The Effects of Lockdown on Air Quality

	AQI		$PM_{2.5}$ ($\mu g/m^3$)	
	levels	log	levels	log
	(1)	(2)	(3)	(4)
>=21 Days Before	-4.29	-0.05	-3.84	-0.04
	(3.18)	(0.03)	(2.72)	(0.03)
14 ~ 21 Days Before	-1.65	-0.03	-1.14	-0.06
	(3.86)	(0.03)	(3.23)	(0.04)
7 ~ 14 Days Before	3.95	0.00	4.20	-0.01
	(3.73)	(0.04)	(3.52)	(0.04)
0 ~ 7 days Later	-9.63***	-0.09***	-7.38***	-0.12***
	(2.62)	(0.03)	(2.23)	(0.03)
7 ~ 14 Days Later	-12.96***	-0.11***	-8.89***	-0.13***
	(2.85)	(0.03)	(2.22)	(0.03)
14 ~ 21 Days Later	-27.69***	-0.28***	-19.65***	-0.33***
	(3.77)	(0.03)	(3.00)	(0.04)
21 ~ 28 Days Later	-29.50***	-0.29***	-20.97***	-0.31***
	(4.57)	(0.04)	(3.65)	(0.05)
>=28 Days Later	-31.43***	-0.31***	-21.50***	-0.38***
	(4.70)	(0.04)	(3.84)	(0.04)
Weather	Y	Y	Y	Y
City FE	Y	Y	Y	Y
Date FE	Y	Y	Y	Y
Obs.	19,764	19,764	19,764	19,764
R-Squared	0.518	0.604	0.527	0.630
No. of Cities	324	324	324	324

Notes: Weather controls include daily temperature, its square, precipitation, and snow depth. Standard errors are clustered at the city level and reported below the coefficients. * significant at 10% ** significant at 5%. *** significant at 1%.

Supplementary Table 6. Event Study: The Effects of 2020 Spring Festival on Air Quality in The Control Group

	AQI		$PM_{2.5}$ ($\mu g/m^3$)	
	levels	log	levels	log
	(1)	(2)	(3)	(4)
>=21 Days Before	5.60	0.03	5.81	0.08
	(4.27)	(0.06)	(3.82)	(0.06)
14 ~ 21 Days Before	0.04	0.02	5.22*	0.04
	(4.74)	(0.04)	(3.07)	(0.05)
7 ~ 14 Days Before	5.80	0.07	2.81	0.02
	(4.36)	(0.05)	(4.22)	(0.06)
0 ~ 7 days Later	2.66	0.02	-7.15	-0.09*
	(4.51)	(0.04)	(4.64)	(0.05)
7 ~ 14 Days Later	-7.72**	-0.04	-2.49	-0.04
	(3.57)	(0.04)	(3.54)	(0.06)
14 ~ 21 Days Later	-10.74**	-0.14***	-10.31***	-0.15***
	(4.66)	(0.05)	(3.43)	(0.05)
21 ~ 28 Days Later	-10.87**	-0.12**	-10.21**	-0.10
	(4.41)	(0.04)	(4.45)	(0.07)
>=28 Days Later	-4.14	-0.07	-3.61	-0.03
	(4.47)	(0.06)	(3.17)	(0.07)
Weather	Y	Y	Y	Y
City FE	Y	Y	Y	Y
Date FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Obs.	27,938	27,938	27,938	27,938
R-Squared	0.485	0.564	0.485	0.577
No. of Cities	229	229	229	229

Notes: Weather controls include daily temperature, its square, precipitation, and snow depth. Standard errors are clustered at the city level and reported below the coefficients. * significant at 10% ** significant at 5%. *** significant at 1%.

Supplementary Table 7. Test for the Differences in Treatment and Control Groups in 2019

	Treatment and Control Groups in 2019			
	Levels		Log	
	(1)	(2)	(3)	(4)
Panel A. Air Quality Index (AQI)				
Spring Festival in 2019	-3.07	-1.08	-0.03	-0.01
* Lockdown cities	(2.29)	(2.07)	(0.03)	(0.03)
R-Squared	0.501	0.518	0.566	0.594
Panel B. $PM_{2.5}$ ($\mu g/m^3$)				
Spring Festival in 2019	-1.61	-0.40	-0.00	0.01
* Lockdown cities	(1.81)	(1.64)	(0.04)	(0.03)
R-Squared	0.513	0.527	0.602	0.623
Weather Control		Y		Y
City FE	Y	Y	Y	Y
Date FE	Y	Y	Y	Y
Obs.	19,764	19,764	19,764	19,764
No. of Cities	324	324	324	324

Notes: Weather controls include daily temperature, its square, precipitation, and snow depth. Standard errors are clustered at the city level and reported below the coefficients. * significant at 10% ** significant at 5%. *** significant at 1%.

Supplementary Table 8. Robustness Check Using Different Subsamples

	AQI		$PM_{2.5}$ ($\mu g/m^3$)	
	Levels	Log	Levels	Log
	(1)	(2)	(3)	(4)
Panel A. Drop Cities in Hubei Province				
Lockdown	-21.80***	-0.19***	-15.70***	-0.18***
	(3.41)	(0.03)	(2.77)	(0.03)
R-Squared	0.516	0.604	0.543	0.644
Obs.	19,032	19,032	19,032	19,032
Weather Control	Y	Y	Y	Y
Date FE	Y	Y	Y	Y
City FE	Y	Y	Y	Y
No. of Cities	312	312	312	312
Panel B. Drop Control Cities Neighboring Lockdown Cities				
Lockdown	-19.91***	-0.17***	-13.10***	-0.14***
	(3.59)	(0.03)	(2.81)	(0.03)
R-Squared	0.515	0.609	0.532	0.661
Obs.	14,152	14,152	14,152	14,152
Weather Control	Y	Y	Y	Y
Week FE	Y	Y	Y	Y
City FE	Y	Y	Y	Y
No. of Cities	232	232	232	232

Notes: Weather controls include temperature, its square, precipitation, and snow depth. In panel B, the weather control variables are aggregated to weekly level. Standard errors are clustered at the city level and reported below the coefficients. * significant at 10% ** significant at 5%. *** significant at 1%.

Supplementary Table 9: The Effect of City Lockdown Before the Spring Festival

	Sample: Before the Spring Festival (~ January 24th)			
	Level		Log	
	(1)	(2)	(3)	(4)
Panel A. Air Quality Index (AQI)				
City Lockdown	-18.45***	-17.87***	-0.13**	-0.11**
	(4.99)	(5.00)	(0.06)	(0.06)
R-Squared	0.633	0.640	0.666	0.681
Panel B. $PM_{2.5}$ ($\mu g/m^3$)				
City Lockdown	-15.78***	-15.41***	-0.13**	-0.12*
	(4.10)	(4.10)	(0.06)	(0.06)
R-Squared	0.630	0.635	0.683	0.693
Weather Control		Y		Y
City FE	Y	Y	Y	Y
Date FE	Y	Y	Y	Y
Obs.	7,776	7,776	7,776	7,776
No. of Cities	324	324	324	324

Notes: Weather controls include daily temperature, its square, precipitation, and snow depth. Standard errors are clustered at the city level and reported below the coefficients. * significant at 10% ** significant at 5%. *** significant at 1%.

Supplementary Table 10. Heterogeneity Analysis

	AQI		$PM_{2.5}$ ($\mu g/m^3$)	
	levels	log	levels	log
	(1)	(2)	(3)	(4)
Panel A				
Cold Region	-25.26***	-0.17***	-14.95***	-0.10***
(Obs. = 9,943)	(5.25)	(0.04)	(3.65)	(0.04)
Warm Region	-6.19***	-0.09***	-4.28***	-0.09***
(Obs. = 9,821)	(1.91)	(0.03)	(1.50)	(0.03)
Northern China	-29.42***	-0.19***	-17.26***	-0.14***
(Obs. = 8,296)	(6.10)	(0.04)	(4.28)	(0.03)
Southern China	-10.02***	-0.12***	-7.86***	-0.10**
(Obs. = 11,468)	(2.87)	(0.03)	(2.39)	(0.04)
Panel B				
GDP (high)	-23.11***	-0.20***	-13.11***	-0.16***
(Obs. = 7,137)	(5.86)	(0.05)	(4.24)	(0.05)
GDP (low)	-18.52***	-0.15***	-14.94***	-0.15***
(Obs. = 12,627)	(3.77)	(0.03)	(3.21)	(0.03)
per capita GDP (high)	-21.56***	-0.19***	-13.04***	-0.15***
(Obs. = 9,455)	(4.65)	(0.04)	(3.49)	(0.04)
per capita GDP (low)	-18.73***	-0.15***	-15.04***	-0.15***
(Obs. = 10,309)	(4.26)	(0.03)	(3.61)	(0.04)
Population (high)	-21.45***	-0.19***	-13.91***	-0.17***
(Obs. = 9,211)	(5.07)	(0.04)	(3.35)	(0.04)
Population (low)	-17.99***	-17.99***	-13.64***	-0.14***
(Obs. = 10,553)	(3.96)	(3.96)	(3.78)	(0.04)
Weather	Y	Y	Y	Y
City FE	Y	Y	Y	Y
Date FE	Y	Y	Y	Y

Notes: Each cell represents a separate regression using the corresponding subsample. For example, a warm region uses cities whose temperature is above the mean. Weather controls include daily temperature, its square, precipitation, and snow depth. Standard errors are clustered at the city level and reported below the coefficients. * significant at 10% ** significant at 5%. *** significant at 1%

Supplementary Table 10. Heterogeneity Analysis (Continued)

	AQI		$PM_{2.5}$ ($\mu g/m^3$)	
	levels	log	levels	log
	(1)	(2)	(3)	(4)
Panel C				
Secondary Industry Output (high)	-31.68***	-0.26***	-20.28***	-0.22***
(Obs. = 7,442)	(5.49)	(0.04)	(3.82)	(0.05)
Secondary Industry Output (low)	-12.53***	-0.11***	-9.71***	-0.11***
(Obs. = 12,322)	(3.75)	(0.03)	(3.27)	(0.04)
No. of Firms (high)	-26.14***	-0.22***	-16.41***	-0.19***
(Obs. = 8,052)	(5.29)	(0.04)	(3.68)	(0.04)
No. of Firms (low)	-14.60***	-0.12***	-11.71***	-0.11***
(Obs. = 11,712)	(4.15)	(0.03)	(3.63)	(0.04)
Amount of Traffic (high)	-23.53***	-0.22***	-13.57***	-0.18***
(Obs. = 7,137)	(5.68)	(0.05)	(4.09)	(0.05)
Amount of Traffic (low)	-17.30***	-0.13***	-13.72***	-0.13***
(Obs. = 12,627)	(3.52)	(0.03)	(3.03)	(0.03)
Panel D				
Wastewater Emission (high)	-21.90***	-0.21***	-13.31***	-0.19***
(Obs. = 8,723)	(4.88)	(0.04)	(3.53)	(0.04)
Wastewater Emission (low)	-18.40***	-0.13***	-14.48***	-0.12***
(Obs. = 11,041)	(4.17)	(0.03)	(3.59)	(0.04)
SO_2 Emission (high)	-23.69***	-0.14***	-14.65***	-0.16***
(Obs. = 8,601)	(5.37)	(0.05)	(4.04)	(0.05)
SO_2 Emission (low)	-16.91***	-0.14***	-13.45***	-0.13***
(Obs. = 11,163)	(3.76)	(0.03)	(3.22)	(0.03)
Dust Emission (high)	-25.03***	-0.22***	-15.90***	-0.18***
(Obs. = 9,394)	(4.89)	(0.04)	(3.66)	(0.04)
Dust Emission (low)	-14.71***	-0.12***	-11.75***	-0.12***
(Obs. = 10,370)	(4.22)	(0.03)	(3.65)	(0.04)
Weather Control	Y	Y	Y	Y
City FE	Y	Y	Y	Y
Date FE	Y	Y	Y	Y

Notes: Each cell represents a separate regression using the corresponding subsample. For example, a warm region uses cities whose temperature is above the mean. Weather controls include daily temperature, its square, precipitation, and snow depth. Standard errors are clustered at the city level and reported below the coefficients. * significant at 10% ** significant at 5%. *** significant at 1%

Supplementary Table 11: Summary of the Environmental Regulations Implemented in China

	Regulations	Description
Panel A. Environmental Regulations		
Chen et al. (2013)[35]	Beijing Olympics	During the Beijing Olympics in 2008, China adopted several radical environmental regulations. These measures led to 30~35 points (28~33%) decrease in the Air Pollution Index.
He et al. (2016)[20]	Beijing Olympics	During the Beijing Olympics, PM_{10} fell around 25 $\mu g/m^3$ (25~27%).
Tanaka et al. (2015)[29]	Two Control Zone	In 1998, China designated nearly 175 prefectures as the Two Control Zones (TCZ), in that local governments were required to reduce SO_2 emissions. In TCZ prefectures, total suspended particulates (TSP) declined by around 60 $\mu g/m^3$ (20%) and SO_2 by 13 $\mu g/m^3$ (15%)
Chen et al. (2018)[30]	Two Control Zone	In TCZ zones, SO_2 emissions declined by around 15%.
Li et al. (2020)[28]	Gasoline Fuel Standard	From 2013 to 2017, China adopted a more stringent gasoline fuel standard. In prefectures, where the new standards were adopted, AQI and $PM_{2.5}$ concentration fell by around 12.9%
Li et al. (2017)[36]	APEC Meeting	During the APEC Meeting in Beijing in 2014 September, China adopted stringent short-term environmental regulations. During the meeting period, they reduced AQI by 37.4%.
Karplus et al. (2018)[37]	Power Plant Emission Standard	China adopted a new emission standard on coal-fired power plants. The regulations are even tighter in some prefectures in the greater Beijing–Tianjin–Hebei area. After the new standard, SO2 emission from the plants declined by 13.9%.
Panel B. Coal Winter Heating System		
Almond et al. (2009)[38]	Huai River Policy	In China, people living in the North (defined by the Huai River) can use a free winter heating system. During winter, TSP concentration is around 204~318 $\mu g/m^3$ (38%~59%) higher in just North of the Huai River than just the South.
Chen et al. (2013)[17]	Huai River Policy	TSP concentration just North of the Huai River is 184 $\mu g/m^3$ (55%) higher
Ebeinstein et al. (2017)[15]	Huai River Policy	PM_{10} concentration just North of the Huai River is 41.7 $\mu g/m^3$ (46%) higher
Fan et al. (2020)[16]	Huai River Policy	Immediately after the start of the central heating system at the beginning of winter, the weekly Air Quality Index increased by 36%

Supplementary References

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